

Assignment 1: Univariate and Multivariate Statistics

FinTech 545 - Quantitative Risk Management

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2026-09-03

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1 Computational conventions

All predictions below are based only on the diagnostic information allowed in the prompt: moments for Problem 1, the scatter plot for Problem 2, pair plots for Problem 3, the covariance matrix for Problem 4, and the series/ACF/PACF for Problem 5. The models were fit only after recording those predictions.

For Problem 1 I compute central moments using denominator n and report excess kurtosis, $m_4/m_2^2 - 3$, matching the course convention. For every model comparison I use

$$\text{AICc} = 2k - 2\ell + \frac{2k^2 + 2k}{n - k - 1},$$

where ℓ is the maximized log likelihood and k counts every fitted parameter, including the error-distribution parameters. Numerical values are rounded in this document, while full precision is available in [results.json](#) and the CSV files in [tables/](#).

2 Reading the Shape of a Sample

2.1 Prediction from the first four moments

For the 1,000 observations in [problem1.csv](#), the first four sample moments are:

Moment	Estimate
Mean	0.00110439
Variance	0.0000961520
Standard deviation	0.00980571
Skewness	-0.668831
Excess kurtosis	2.339849

The sample is materially left-skewed and leptokurtic. Using only skewness and excess kurtosis:

- **Normal is ruled out.** A Normal has skewness 0 and excess kurtosis 0, so both shape moments disagree with the sample.
- **Lognormal is ruled out.** A non-degenerate Lognormal has positive skewness, while this sample has negative skewness. This conclusion does not require using the support of the data.
- **Symmetric Student's t is ruled out.** It can produce positive excess kurtosis, but its skewness is 0 whenever the skewness exists.
- **Normal inverse Gaussian (NIG) remains plausible.** Its skew parameter can produce negative skewness and it can also produce positive excess kurtosis.

This is a screening conclusion, not proof that the NIG generated the sample; the first four moments do not uniquely identify a distribution.

2.2 Normal fit and tail count

I matched the Normal mean and variance to the sample. Therefore

$$X_{\text{Normal}} \sim N(0.00110439, 0.0000961520).$$

Its 1% quantile is **-0.0217071**. There are **26 observations** below that threshold, whereas a correctly calibrated 1% quantile would produce $1000(0.01) = \mathbf{10}$ observations on average.

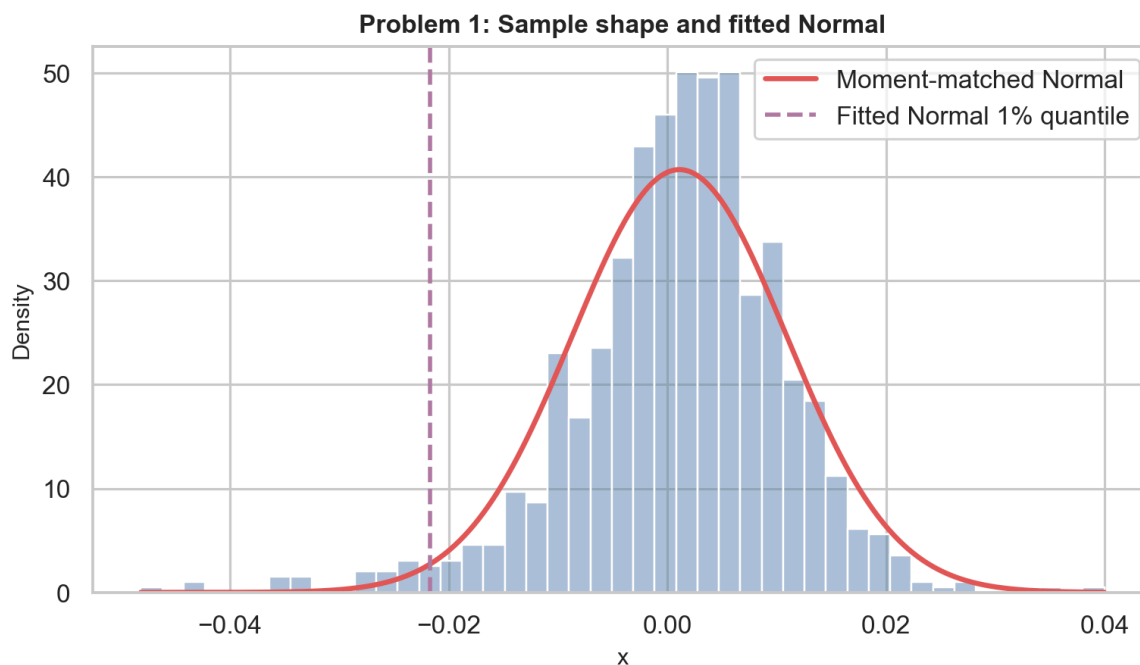


Figure 1: The fitted Normal misses the sample's asymmetry and tail shape. The dashed line is the fitted Normal's 1% quantile.

2.3 Reconciliation

The fit confirms the moment-based warning. The observed lower-tail count is 2.6 times the Normal expectation. A symmetric Normal cannot allocate extra mass specifically to the left tail, and zero excess kurtosis prevents it from matching the sample's combination of a sharp center and heavy tails. Thus the Normal understates the probability and severity of large negative outcomes. For risk measurement this error is in the dangerous direction: a Normal-based left-tail loss threshold would be too optimistic.

3 A Regression Whose Errors Are Not Normal

3.1 Prediction from the scatter plot

The scatter is centered on a strong, approximately linear relation with slope near 3 and intercept near 1.5. Most observations form a tight band, but several vertical deviations are far larger than a Normal cloud would suggest. The spread looks broadly symmetric rather than one-sided, and there is no dominant fan pattern. I therefore predict a **symmetric Student's t error** with few degrees of freedom rather than a Normal error.

This violates **OLS assumption 7: the error term is normally distributed**. If assumptions 1–6 continue to hold, non-Normality does not create a systematic bias in the OLS slope: OLS is still unbiased/consistent, although it is more sensitive to extreme observations and is no longer the efficient likelihood estimator for a t error. Separately, the usual residual-variance formula can still estimate the slope's conditional variance under independent, homoskedastic finite-variance errors, but the exact small-sample Student- t reference distribution for tests and intervals no longer follows. Heavy outliers can also inflate the realized residual variance and make the reported standard error unstable. I therefore expect the estimated slope to move only modestly, while Normal-based inference and tail probabilities are less trustworthy.

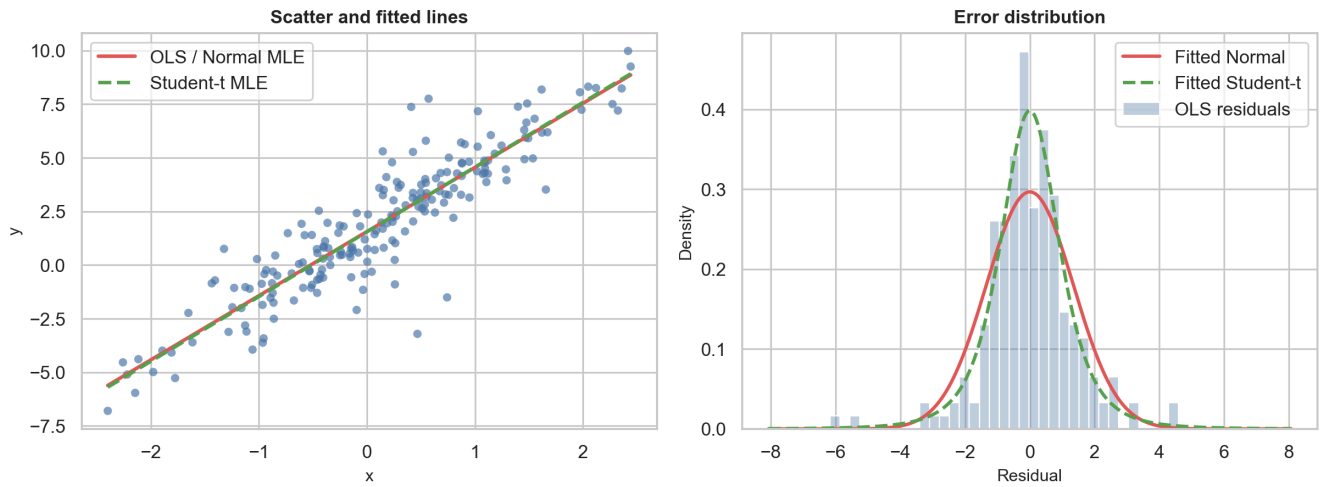


Figure 2: The fitted lines are nearly identical, while the residual panel shows a sharper center and heavier tails than the fitted Normal.

3.2 Fits and AICc

The fitted values are:

Model	$\hat{\alpha}$	$\hat{\beta}$	Error scale	ν	AICc
OLS	1.560740	2.990757	1.351401	–	692.145*
Normal MLE	1.560740	2.990757	1.344627	–	692.145
Student- <i>t</i> MLE	1.547979	3.019179	0.935509	3.632163	665.327

For OLS, the standard errors are **0.096269** for $\hat{\alpha}$ and **0.096640** for $\hat{\beta}$. Its error scale is the residual standard error $\sqrt{\text{SSE}/(n-2)}$. The Normal MLE uses $\sqrt{\text{SSE}/n}$, which explains the small scale difference. Normal MLE and OLS have identical coefficient estimates because maximizing a Normal regression likelihood over α and β is equivalent to minimizing SSE.

The Student-*t* scale parameter is not its standard deviation. With $\hat{\nu} = 3.632163$, its implied standard deviation is 1.395562. The asterisk on the OLS AICc indicates that distribution-free OLS has no likelihood and hence no AICc of its own; the displayed value is obtained only after attaching the fitted Normal likelihood, making it the Normal MLE comparison.

Student-*t* improves AICc by **26.817** relative to Normal even after paying for one additional parameter, so the Student-*t* error model is strongly preferred.

3.3 Reconciliation of slopes and error shape

The OLS/Normal slope is 2.990757 and the Student-*t* slope is 3.019179, a difference of only 0.028422 (about 0.95% of the OLS slope). The violation did not move the slope much, consistent with the prediction that the conditional mean is well specified and the main problem is the error distribution.

The rejected Normal model gets the **shape of the residual distribution** wrong. To accommodate extreme errors with one scale parameter, it spreads too much probability through the shoulders of the distribution, yet it still assigns too little probability to the far tails and too little density to the sharp central peak. This is why the likelihood strongly favors Student-*t* even though the fitted regression lines are visually almost identical.

3.4 Error quantiles and a capital buffer

Upper-tail probability	Normal error quantile	Student- t error quantile	Wider model
95.0%	2.211715	2.053833	Normal
99.5%	3.463530	4.620086	Student- t

At 95%, the smaller fitted t scale dominates, so the Student- t quantile is narrower. Farther into the tail, its power-law tail dominates and the 99.5% quantile becomes much wider than the Normal quantile. For a capital buffer I would use the **Student- t model**, because a buffer is designed for rare, severe outcomes, precisely where the Normal fit is too thin-tailed. The same magnitudes apply to the lower tail with the sign reversed because both fitted error distributions are symmetric.

4 Pearson Against Spearman

4.1 Prediction from all pair plots

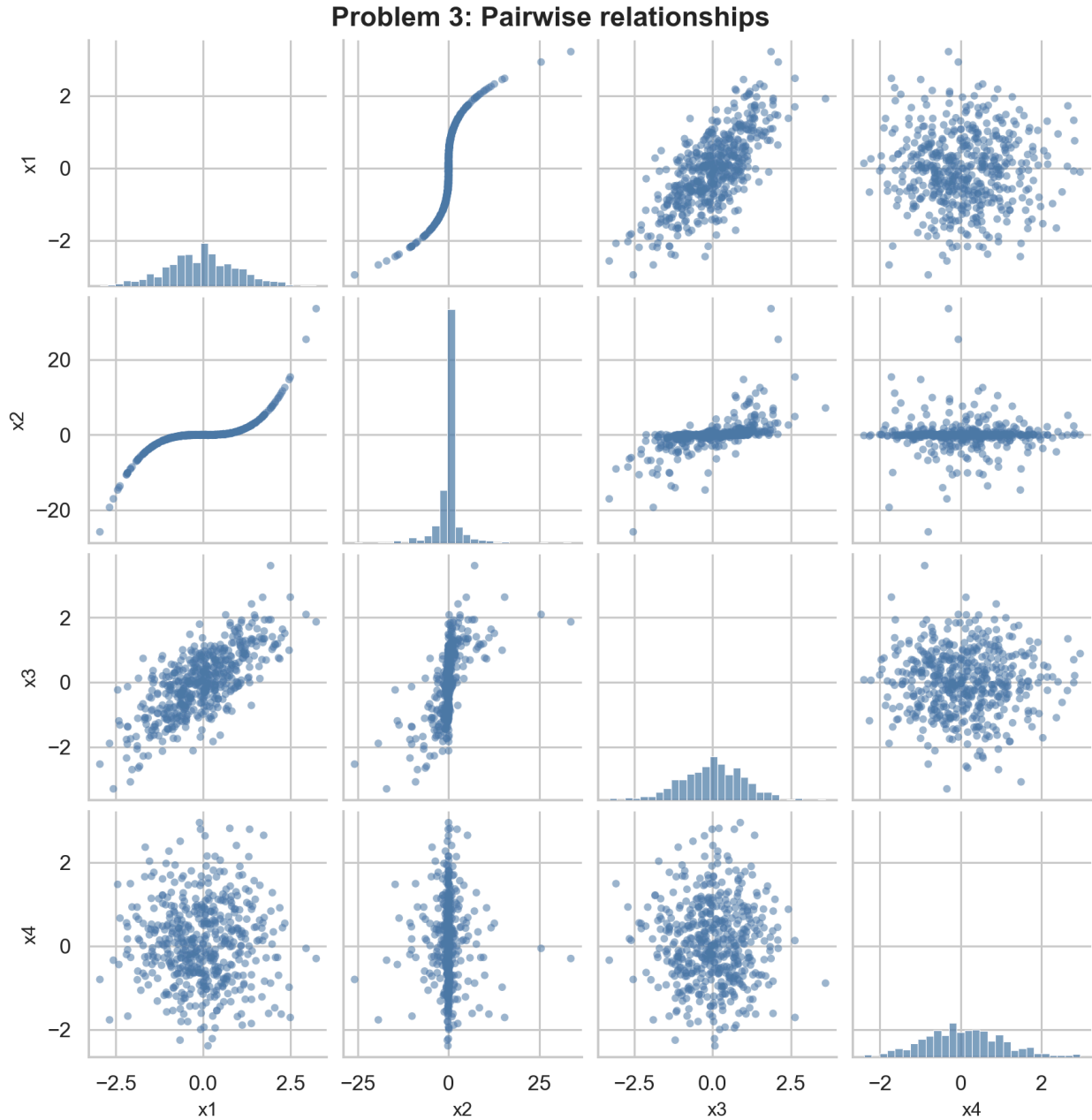


Figure 3: Pair plots reveal a strong but curved monotone relation between x_1 and x_2 , an approximately linear relation between x_1 and x_3 , and little relation involving x_4 .

The x_1 - x_2 pair should produce the largest disagreement. Its relation is strongly monotone but visibly S-shaped/curved, so the ranks are nearly preserved even though a straight line is not an exact summary. I expect Spearman to be much higher than Pearson.

I expect the measures to agree most closely for every pair involving x_4 , because those clouds show no systematic relation and both measures should be near zero. They should also be reasonably close for x_1 - x_3 , whose cloud is approximately linear. The x_2 - x_3 relation is monotone but inherits curvature from x_2 , so I expect a smaller version of the x_1 - x_2 gap.

4.2 Correlation matrices

Pearson correlation:

	x_1	x_2	x_3	x_4
x_1	1.000	0.799	0.719	-0.005
x_2	0.799	1.000	0.588	-0.021
x_3	0.719	0.588	1.000	-0.009
x_4	-0.005	-0.021	-0.009	1.000

Spearman correlation:

	x_1	x_2	x_3	x_4
x_1	1.000	0.972	0.684	0.000
x_2	0.972	1.000	0.667	-0.001
x_3	0.684	0.667	1.000	-0.002
x_4	0.000	-0.001	-0.002	1.000

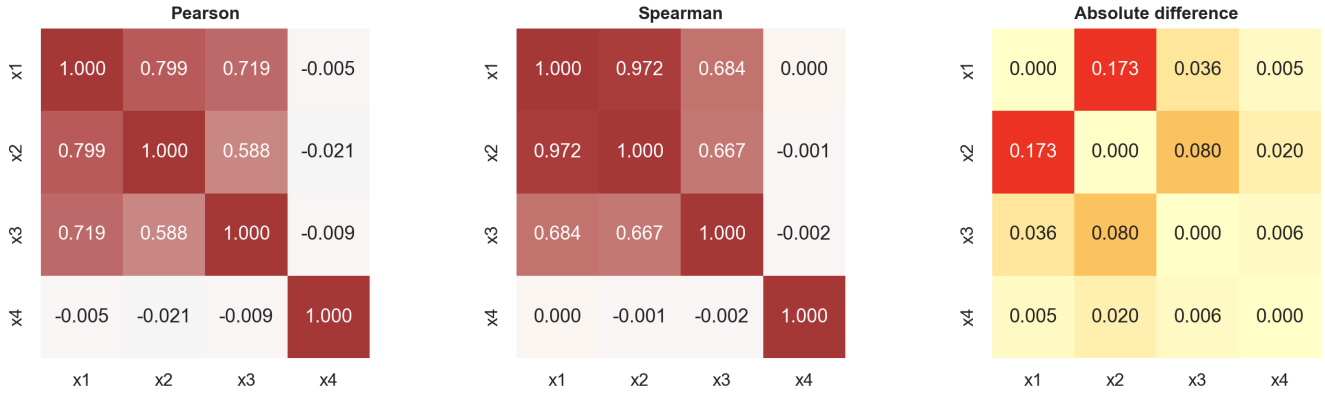


Figure 4: Pearson, Spearman, and their absolute difference.

The largest absolute gap is for x_1 and x_2 :

$$| 0.799092 - 0.971930 | = 0.172838.$$

4.3 Reconciliation and mechanism

This matches the prediction. Pearson uses the original distances from the means and measures **linear** co-movement. The strong curvature means one straight-line slope cannot describe the entire $x_1 - x_2$ relation, and the large-magnitude x_2 observations receive substantial leverage. Spearman first replaces values by ranks, discarding the nonlinear spacing while preserving the nearly monotone ordering.

Spearman's 0.972 is the more honest answer to "how strongly do these variables move monotonically together?" Pearson's 0.799 answers a different, still useful question: "how much standardized linear covariance do these variables share?" For a covariance matrix or a linear hedge, Pearson is the relevant quantity; for the strength of this nonlinear dependence, Spearman is more descriptive.

5 Conditional Distributions

5.1 Prediction from the covariance matrix

The sample mean vector and covariance matrix are

$$\hat{\mu} = \begin{bmatrix} -0.016714 \\ 0.045288 \end{bmatrix}, \quad \hat{\Sigma} = \begin{bmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{bmatrix}.$$

I define $\Sigma_{11} = \text{Var}(x_1)$, $\Sigma_{22} = \text{Var}(x_2)$, $\Sigma_{12} = \text{Cov}(x_1, x_2)$, and $\Sigma_{21} = \Sigma'_{12}$. Thus x_1 is the observed/conditioning block and x_2 is the target block. Under the multivariate Normal,

$$\text{Var}(x_2 | x_1) = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}.$$

The conditional-to-unconditional variance factor is

$$\frac{\Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}}{\Sigma_{22}} = 1 - \frac{\Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}}{\Sigma_{22}} = \mathbf{0.362201}.$$

Learning x_1 therefore reduces the variance of x_2 to 36.22% of its unconditional value, a 63.78% reduction. In standard-deviation units the factor is $\sqrt{0.362201} = 0.601832$.

Under the multivariate Normal, this factor **does not depend on the observed value of x_1** . The conditional variance formula contains only covariance blocks; unlike the conditional mean formula, it contains no term involving the realized x_1 .

5.2 Conditional mean, band, and coverage

The conditional mean is

$$E[x_2 | x_1] = \mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (x_1 - \mu_1).$$

For this sample,

$$E[\widehat{x_2} | x_1] = 0.045288 + 1.542813(x_1 + 0.016714) = \mathbf{0.071073 + 1.542813x_1}.$$

The coefficient $\Sigma_{21} \Sigma_{11}^{-1} = 1.542813$ is exactly the OLS slope from regressing x_2 on x_1 with an intercept. The fitted conditional variance is 1.486746, so the conditional standard deviation is 1.219322. I drew the 95% band as the fitted conditional mean plus or minus 1.959964(1.219322).

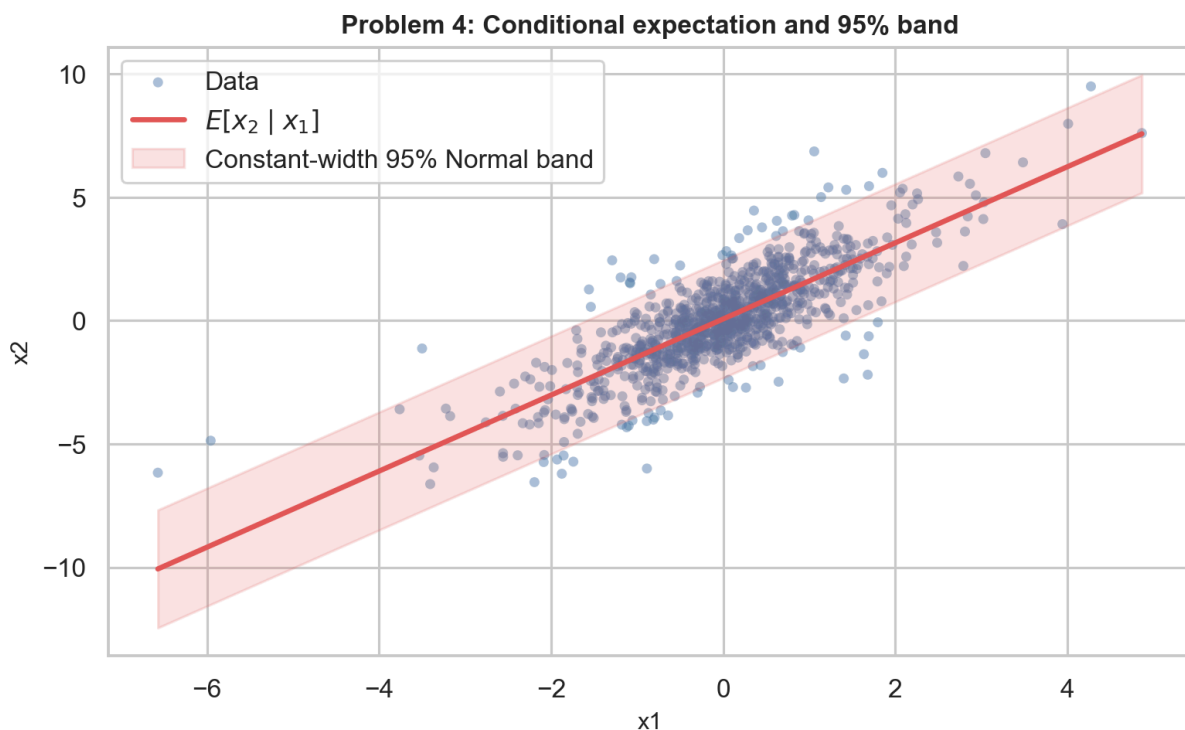


Figure 5: Conditional expectation and the constant-width Normal 95% band.

Overall, **935 of 1,000 observations**, or **93.5%**, fall inside the band.

Distance of x_1 from its mean	Observations	Inside	Coverage	Residual SD
Within 1 SD	759	726	95.65%	1.0758
Between 1 and 2 SD	190	163	85.79%	1.5804
Beyond 2 SD	51	46	90.20%	1.5957

5.3 Reconciliation and failed assumption

Coverage is close to nominal near the center but substantially lower away from the center. The residual standard deviation rises from about 1.08 in the central bucket to about 1.58–1.60 in the two outer buckets. Thus the data are **heteroskedastic**: $\text{Var}(x_2 | x_1)$ changes with the location of x_1 . This violates the multivariate-Normal implication of a constant conditional covariance, so a single constant-width band cannot be calibrated equally across the range of x_1 .

Some parts of the conditional-mean calculation survive. The coefficient $\Sigma_{21}\Sigma_{11}^{-1}$ remains the population/sample **best linear projection coefficient** without joint Normality, and the plotted line remains a useful linear summary if the conditional mean is in fact linear. What no longer follows automatically is that this line is the exact conditional expectation at every x_1 . More clearly, the constant conditional variance, the constant band width, the exact Normal conditional distribution, and exact 95% coverage do not survive. The 0.3622 variance factor should therefore be interpreted as a pooled linear-projection variance reduction rather than a universal conditional-uncertainty factor for every observed x_1 .

6 Identifying an AR or MA Order

6.1 Prediction from the series, ACF, and PACF

The series fluctuates around a stable mean near 0.67 without a trend. Using the course significance band,

$$\pm \frac{1.96}{\sqrt{500}} = \pm 0.087654.$$

The ACF has a large positive lag 1 (0.462), is near zero at lag 2, becomes negative at lags 3–4, and later returns toward positive values. That is a damped, oscillating **decay**, not a clean cutoff. The PACF has decisive spikes at lags 1 (0.462) and 2 (-0.303), then is mostly inside the band. I therefore predict an **AR(2)** process: for $AR(p)$, the ACF decays and the PACF cuts off after lag p ; for $MA(q)$, those roles reverse.

There is an isolated PACF exceedance at lag 9. I do not treat one late spike as a new cutoff: when 20 lags are compared with individual 5% bands, roughly one false exceedance is expected by chance even if the later true PACFs are zero.

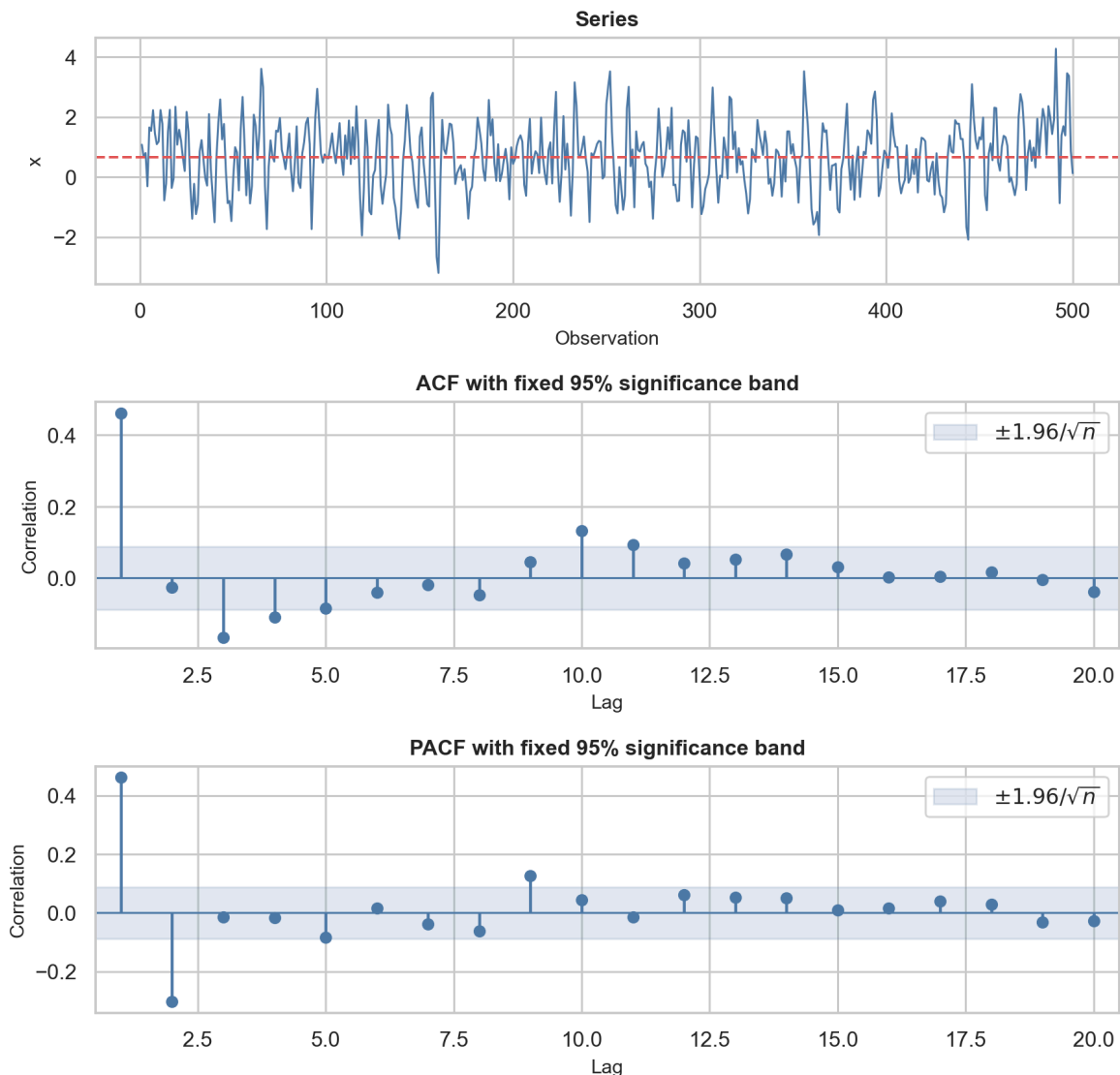


Figure 6: The ACF decays with an oscillating pattern while the PACF's structural spikes are concentrated at lags 1 and 2.

6.2 Model fits and AICc selection

Every model includes a constant and a Gaussian innovation variance. The exact state-space maximum-likelihood fits give:

Model	k	Log likelihood	AICc
AR(1)	3	-706.2693	1418.5870
AR(2)	4	-682.1788	1372.4383
AR(3)	5	-682.1109	1374.3432
MA(1)	3	-691.5179	1389.0843
MA(2)	4	-686.5754	1381.2316
MA(3)	5	-681.9975	1374.1165

AICc selects **AR(2)**, matching the prediction. Its fitted model is

$$(x_t - 0.670607) = 0.601201(x_{t-1} - 0.670607) - 0.302638(x_{t-2} - 0.670607) + \epsilon_t,$$

with innovation variance 0.895802.

6.3 AR(2) versus AR(3)

AR(3) estimates coefficients 0.596203, -0.292690, and **-0.016542**. The third coefficient is very small. Adding it improves the log likelihood by only $-682.1109 - (-682.1788) = 0.0679$, while it adds one fitted parameter. The tiny fit gain cannot repay AICc's complexity penalty, so AICc rises from 1372.4383 to 1374.3432 and rejects AR(3).

Ordinary R^2 would not make the same tradeoff. Adding another lag enlarges the regressor set, so in-sample SSE cannot increase and R^2 cannot decrease, even when the added coefficient is fitting noise. AICc explicitly charges for the extra parameter and therefore favors the more parsimonious AR(2).

7 Reproducibility

All calculations and figures are generated by `analysis.py`. The accompanying [README.md](#) gives the exact environment setup and commands needed to reproduce every reported result and rebuild this PDF.